MIL-I-46231B(MU)
27 SEPTEMBER 1963

SUPERSEDING
MIL-I-46231A(MU)
13 MAY 1963

MILITARY SPECIFICATION

IGNITER, ROCKET MOTOR, T21E2 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the quality assurance provisions, and special requirements not covered by the drawing, for loading, assembling and packing one type of initiating device designated as Igniter, Rocket Motor, T21E2 (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-I-45208 — Inspection Requirements, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8799887 — Igniter, Rocket Motor, T21E2, Assembly.

8839440 — Box, Packing, Ammunition, for Igniter, Rocket Motor, T21E2.

8839441 — Carton Packing, Ammunition, for Igniter, Rocket Motor, T21E2.

PUBLICATIONS

ORDNANCE CORPS

IEL-8799887 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1340

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Igniter assemblies. The igniter assemblies shall comply with all requirements specified on Drawing (dwg.) 8799887 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder. The moisture content of the black powder at the loading plant at the time of loading shall not exceed 0.60 percent.

3.4 Leakage in the assembly. The igniter assembly, when submerged in water a minimum of 1 inch, shall not exhibit leakage, as evidenced by the escape of air into the water, when submitted to an externally applied vacuum of 10 inches of mercury, minimum, applied for 15 seconds, minimum. The water shall be maintained at 70 plus or minus 5 degrees Fahrenheit (F.).

3.5 Functioning. The igniter assembly, after temperature conditioning at minus 65 plus or minus 2 degrees F., shall function with an ignition delay time of 100 milliseconds, maximum, when 2 amperes, plus or minus 5 milliamperes, is applied between the leads. (Ignition delay time is defined as the period between the application of the current and unit functioning.)

3.6 Workmanship. All parts and assemblies shall be free of surface defects, dirt, grease, corrosion products and other foreign matter. The cleaning methods used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 25 assemblies and 25 cover-squib subassemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 *Inspections to be performed.* Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3.1 of this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly or component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample, or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of a shipment of igniter assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot interfix number. Each lot shall contain:

(a) Inert parts from lots bearing the same lot interfix number from one manufacturer.

(b) Squibs from not more than one lot.

(c) Black powder from not more than one lot.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans

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may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Assembly, prior to loading (see dwg. 8799887).

Categories	*Defects*	*Method of inspection*	*Code No. (see 4.2)*
Critical:			
1.	Squib leads not shorted or shorting clip improperly assembled	Visual	01001
Major:	AQL 0.40 percent		
101.	Weight of charge	Balance	01002
102.	Foreign matter on interior surface	Visual	01003
103.	Squib improperly connected or positioned	Visual	01004
104.	Squib damaged	Visual	01005
Minor:	None defined		

4.3.2.2 Assembly (see dwg. 8799887).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Charge missing	Visual	02001
102.	Lead wire damaged	Visual	02002
103.	Insulation on lead missing or damaged	Visual	02003
104.	Cup or cover damaged	Visual	02004
Minor:	AQL 1.50 percent		
201.	Total height, maximum (max.)	Gage	02005
202.	Excess flash at heat seal	Visual	02006
203.	Evidence of poor workmanship (see 3.5)	Visual	02007

4.3.2.3 Carton, sealed (see dwg. 8839441).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Contents loose	Manual	03001
102.	Sealing strip insecure	Visual	03002
103.	Corner not blunted	Visual	03003
Minor:	AQL 0.40 percent		
201.	Marking missing, misleading or unidentifiable	Visual	03004

4.3.2.4 Barrier bag, sealed (see dwg. 8839441).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Air not exhausted	Visual	04001
102.	No room for subsequent seals	Visual	04002
103.	Seal inadequate	Visual-Manual	04003
Minor:	AQL 0.40 percent		
201.	Marking missing, misleading or unidentifiable	Visual	04004

4.3.2.5 Box, sealed (see dwg. 8839440).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Contents loose (when tilted)	Manual	05001
102.	Board broken or contents exposed	Visual	05002
103.	FSC number missing or misleading	Visual	05003
104.	Strapping missing, broken or loose	Visual-Manual	05004
105.	Hardware missing, broken or loose	Visual-Manual	05005
Minor:	AQL 0.40 percent		
201.	Marking missing, misleading or unidentifiable	Visual	05006

4.3.3 *Testing.*

4.3.3.1 Initial production sample (see 4.2.1). The sample submitted as the initial production quantity shall be tested as specified below after having been examined for any or all other drawing requirements. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Strength of lead wire crimp (see dwg. 8799887)—Code No. 02008.

4.3.3.1.2 Electrical resistance (see dwg. 8799887)—Code No. 02009.

4.3.3.1.3 Moisture content of black powder (see 3.3)—Code No. 06001.

4.3.3.1.4 Leakage testing (see 3.4)—Code No. 07001.

4.3.3.1.5 Functioning (see 3.5)—Code No. 07002.

4.3.3.2 *Production testing.*

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4.3.3.2.1 Strength of lead wire crimp (see dwg. 8799887)—Major defect—Code No. 02008. A sample of 35 cover-squib subassemblies from each lot shall be submitted to this test. Lead wire slippage in one or both connectors on one or more units of the sample or damaged insulation shall reject the lot subject to 100 percent testing. During 100 percent testing failure of either requirement shall reject that unit subject to rework and resubmission to test. Testing shall be as specified in 4.4.1 using the applicable test equipment of 4.3.4.

4.3.3.2.2 Electrical resistance (see dwg. 8799887)—Major defect—Code No. 02009. This determination shall be performed 100 percent as specified in 4.4.2 using the applicable test equipment of 4.3.4. Any igniter assembly failing to meet the requirement shall be classed defective and be removed from the lot.

4.3.3.2.3 Moisture content of black powder (see 3.3)—Major defect—Code No. 06001. The black powder from 3 igniter assemblies selected from each lot shall be submitted to the determination of 4.4.3 and the results averaged. If the average exceeds the percentage permitted the lot shall be rejected.

4.3.3.2.4 Leakage testing (see 3.4)—Major defect—Code No. 07001. This test shall be performed 100 percent using the applicable test equipment of 4.3.4 as specified in 4.4.4. Any unit exhibiting leakage shall be classed defective and removed from the lot.

4.3.3.2.5 Functioning (see 3.5)—Major defect—Code No. 07002. A sample of 35 igniter assemblies shall be selected from each lot and be subjected to this test. Testing shall be as specified in 4.4.5 using the applicable equipment of 4.3.4. Failure of one or more assemblies in meeting either requirement shall reject the lot.

4.3.4 *Inspection equipment.* Index of inspection equipment number IEL-8799887 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Strength of lead wire crimp.* The lead wires on both sides of the connector shall be gripped and the required force applied parallel to the center line of the connector. The dwell time shall be 5 seconds, minimum. Observation shall then be made at each end of the connector for lead wire slippage and for damaged insulation. Any unit exhibiting wire slippage in either connector or insulation damage shall be classed defective.

4.4.2 *Electrical resistance.* The short shall be removed from the lead wires of the assembly and the lead wires connected to the test equipment. (The equipment shall limit the current to 10 milliamperes, maximum). Any unit exhibiting a resistance outside the permitted range specified on the applicable drawing shall be classed defective. The lead wires shall be shorted immediately after test.

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Warning: **This test may be dangerous and should be performed behind a suitable barricade.**

4.4.3 *Moisture content of black powder.* The black powder from each assembly shall be accurately weighed in tared covered weighing dishes. The dishes, with covers removed, shall be placed in an oven for 4 hours. The temperature in the oven shall be maintained at 70 degrees Centigrade (C.) to 75 degrees C. The dishes shall then be covered, removed and cooled in a desiccator and weighed. The loss in weight shall be calculated as percent moisture. The 3 determinations shall be averaged.

4.4.4 *Leakage testing.* The igniter shall be placed in the vacuum chamber and submerged to the required depth. (The lead wire ends need not be submerged.) The chamber shall be closed and the required vacuum applied and held for the specified period. During that period observation shall be made for leaks. Any unit exhibiting leakage shall be classed defective.

4.4.5 *Functioning.* The igniter shall be conditioned in a conditioning chamber at the specified temperature, plus or minus 5 degrees F., for a period of 16 hours, minimum. During the same period an insulating container shall be temperature conditioned. (A closed cell, light weight foamed plastic container ¼ to ½ inch thick is suggested. The clearance between the cover and the container shall be such that the lead insulation is not damaged.) While in the conditioning chamber the igniter shall be placed in the container with the leads protruding outside the container and the cover shall be assembled. The leads shall be untwisted and connected to the test equipment, the cover removed, the igniter taken out and shall be fired within 15 seconds, using the specified current. The ignition delay shall be observed and recorded. Any unit which fails to function or which exceeds the ignition delay time permitted shall be classed defective.

Note. The igniter may be fired while enclosed in the container.

Caution: **This test may be dangerous and should be performed behind a suitable barricade.**

5. PREPARATION FOR DELIVERY

5.1 Packaging and marking.

5.1.1 *Level A.* The loaded igniter assemblies shall be packaged and marked in accordance with dwg. 8839441.

5.2 Packing and marking.

5.2.1 *Level A.* The packaged igniters shall be packed and marked in accordance with dwg. 8839440.

5.3 Data cards. Data card information shall be as specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF — Contractor's design responsibility on Ordnance format; ORDM-608-12.

CDCF — Contractor's design responsibility on contractor format.

OD — Ordnance design.

ODMU — Ordnance design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608-12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 *Submission of initial production sample.* Instructions, as to the location for the evaluation

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of the initial production sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

6.5 Intended use. It is intended that the item of this specification will be used in Rocket Motor, M95.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Preparing activity:

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